

Three-dimensional linear transformations

CHAPTER 5

Three-dimensional linear transformations

The same method with one more component.

THE IDEA

Everything from 2D carries over. There is now a third basis vector $\mathbf{k} = (0, 0, 1)$ alongside $\mathbf{i}$ and $\mathbf{j}$, so matrices become 3×3 — three columns, one for where each basis vector lands.

2D	3D
$\mathbf{i}, \mathbf{j}$	$\mathbf{i}, \mathbf{j}, \mathbf{k}$
2x2 matrix, 2 columns	3x3 matrix, 3 columns
$x \cdot (\text{col1}) + y \cdot (\text{col2})$	$x \cdot (\text{col1}) + y \cdot (\text{col2}) + z \cdot (\text{col3})$

No new concept — only one extra term to track. Composition works identically: push each basis vector through both matrices.

WORKED EXAMPLE

CHAPTER 5 — WORKED EXAMPLE (3D)

same method, one extra component ($\hat{k}$)

RULE

$$(x,y,z) \rightarrow x \cdot (\text{col 1}) + y \cdot (\text{col 2}) + z \cdot (\text{col 3})$$

EXAMPLE

$$\hat{i} \rightarrow (2,0,1) \quad \hat{j} \rightarrow (0,3,0) \quad \hat{k} \rightarrow (-1,1,2)$$

input (1, 2, -1)

$$1 \times (2, 0, 1) = (2, 0, 1)$$

$$2 \times (0, 3, 0) = (0, 6, 0)$$

$$-1 \times (-1, 1, 2) = (1, -1, -2)$$

add each component :

$$x : 2 + 0 + 1 = 3$$

$$y : 0 + 6 + (-1) = 5$$

$$z : 1 + 0 + (-2) = -1$$

$$\text{result} = (3, 5, -1)$$

Why it matters: real machine-learning data lives in hundreds or thousands of dimensions. You cannot picture them, but the arithmetic never changes — just more components.

FORMULAS FROM THIS CHAPTER

Basis in 3D

$\hat{i}, \hat{j}, \hat{k} \rightarrow 3 \times 3$ matrix, 3 columns

Apply to vector

$x \cdot (\text{col 1}) + y \cdot (\text{col 2}) + z \cdot (\text{col 3})$